

Change-Point Tests And Audit-Discrepancy Clustering For RCOUNT Packages: A Method Contract For Investigative, Non-Certifying Sequence Analytics

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Abstract

Change-point tests and audit-discrepancy clustering are investigative election forensic methods. They can prioritize review by flagging abrupt shifts in reported-result sequences and by asking whether audit discrepancies concentrate by scanner, batch, or source. They do not prove fraud, certify correctness, or replace risk-limiting audits. This paper instantiates the W.01 forensic report contract for two sequence-oriented analytic families: change-point detection on RCOUNT status-event and batch-order series, and concentration tests over V-series audit observation discrepancies. The contribution is a method framework and report shape, not shipped software: no forensics crate, code fixture, function, or test is claimed here. The methods are established, the contract is defined in W.01, and the fixtures described below are intended roles for later implementation.

1 Introduction

Forensic analytics can prioritize investigation. They cannot prove fraud, certify correctness, or replace risk-limiting audits. This boundary is not a footnote for the W-series; it is the first claim every method must make. A score from this paper is a reason to ask a question, not an answer to that question.

This method paper covers two time- and sequence-oriented signals. The first is an abrupt shift in a reported-results series: turnout, vote share, margin, residuals, or other normalized features ordered by reporting time or by batch sequence. Such a shift may be a change point worth review. The second is a set of audit discrepancies that cluster by source, scanner, batch, or similar operational unit instead of scattering as the null model would expect.

Both signals are useful because RCOUNT packages preserve the record structure needed to replay them: status events, batch manifests, source hashes, and audit observations. Both are also easy to overread. Mail batches reported first, late-arriving provisional ballots, one high-volume scanner, or a local reporting workflow can create legitimate discontinuities and clusters. The purpose here is therefore narrow: instantiate the W.01 forensic report contract for these two families and keep the result investigative, reproducible, and separate from certification evidence.

2 Method

2.1 Change-Point Tests

A change-point test takes a reported-results series ordered by time or batch position and asks whether one or more locations split the sequence into regimes with meaningfully different behavior. The feature may be a cumulative margin, per-batch vote share, turnout rate, residual from a comparison model, or another normalized RCOUNT-derived quantity. Candidate statistics include CUSUM scores, segmentation costs, and likelihood-ratio change-point statistics.

The statistic is not interpreted alone. The report must record the ordering rule, the feature set, the baseline population, and the operational note that reporting order often creates benign changes. For example, mail ballots or early-vote batches may be reported before election-day precincts; central-count batches may arrive in blocks; small late batches may have different composition than earlier high-volume batches. A flagged change point is therefore an investigation item: it identifies the segment boundary and the observed shift, then asks whether the reporting workflow explains it.

2.2 Audit-Discrepancy Clustering

A discrepancy-clustering test takes audit observations from the V-series side of an RCOUNT package and asks whether discrepancies concentrate in operational groups. The grouping variable may be scanner, batch, source file, tabulator, audit board, or another unit preserved by the package. The null is random scatter conditional on the relevant audit design and group sizes. Candidate statistics include chi-square tests over grouped counts, concentration indices, maximum-cell overrepresentation scores, and permutation or rank-based clustering statistics.

The method consumes discrepancies as observations, not as certification findings. It asks whether the pattern of discrepancies is more concentrated than expected under the selected random-scatter baseline. A high score should name the grouping unit, the observed concentration, the comparison population, and the caveat. It should not say that the scanner, batch, or source caused the problem, and it should not convert an audit discrepancy into proof of outcome error or fraud.

3 RCOUNT Inputs And Report Contract

Change-point tests consume RCOUNT status events and batches. Status events give the reported-results timeline; batch manifests give an alternate sequence order when the analytic question is about tabulation or upload batches rather than wall-clock time. The scored unit is usually a time segment, batch interval, or candidate boundary between adjacent segments.

Discrepancy-clustering tests consume V-series audit observations and derived discrepancy records. The scored unit is usually a scanner, batch, source, or other operational grouping. This is the point where W-series analytics may use V-series outputs, but only as inputs to an investigative report. The W-series artifact must not recast those observations as certification evidence.

Both methods instantiate the W.01 report contract, which requires source hashes, feature set, baseline population, model identity, unit-level scores, caveats, and an investigative-not-certifying statement. For these two families the per-record transcript is:

Field	Instantiation
<code>method_id</code>	Registered W.06 analytic: <code>change-point</code> or <code>discrepancy-cluster</code> .
<code>feature_set</code>	Time/batch reporting features, or V-series discrepancy fields and grouping variables.
<code>baseline_population</code>	Comparison window, segment set, audit-observation universe, and grouping universe.
<code>model_id</code>	<code>change-point</code> or <code>discrepancy-cluster</code> .
<code>unit_id</code>	<code>time-segment</code> for sequence boundaries, or scanner, batch, or source for discrepancy groups.
<code>score</code>	Change-point statistic, segmentation score, concentration statistic, or clustering score.
<code>p_value_or_rank</code>	Optional calibration or rank under the stated baseline.
<code>investigation_note</code>	Caveat describing reporting order, operational grouping, or audit-design constraint.

The report also binds the source package hashes used to compute the features. That binding is part of the method: another analyst with the same package, feature definition, and model identity should be able to reproduce the same score table. This paper defines that contract instantiation; it does not claim that a forensics crate, function, fixture, or test already exists.

4 Intended Fixtures

The fixtures for this paper follow the four-role W-series pattern defined by W.01. They are intended fixtures for later implementation, not existing code. Each method family needs the same roles with method-specific content.

Role	Change-point fixture	Discrepancy-clustering fixture
No anomaly	A stable reporting sequence with no planted regime shift.	Audit discrepancies, if any, scattered according to the stated random baseline.
Injected anomaly	A planted abrupt change in the ordered series.	A planted concentration of discrepancies in one scanner, batch, or source.
Innocent structure	Mail-first or otherwise grouped reporting that creates a legitimate shift.	One high-volume scanner or batch with more observations because it processed more ballots.
False-positive boundary	A high statistic whose investigation note points to the reporting workflow.	A high concentration score whose note points to operational volume or audit design.

The no-anomaly role checks that the method does not manufacture an investigation queue from ordinary variation. The injected-anomaly role checks that the statistic can surface the planted signal. The innocent-structure role checks that the feature design preserves known operational explanations. The false-positive boundary role is the most important: it verifies that the report can carry a high score and still say, plainly, that the score is not proof.

5 Boundaries

Reporting order and operational structure routinely create legitimate change points and clusters. Mail ballots may be reported before precinct ballots; central-count batches may be uploaded together; recount or adjudication events may arrive late; scanners and batches may differ greatly in volume. A method that ignores those facts will produce impressive-looking false positives.

Consuming V-series audit discrepancies does not change the W-series claim. The V-series records are evidence-bound audit observations. A W-series clustering report can ask whether their distribution is unusual, but it cannot promote that unusual distribution into proof of fraud, a certification result, or a replacement for a risk-limiting audit. The transcript kinds must remain separate.

The methods are descriptive, not causal. They identify where a sequence changes or where discrepancies concentrate. The reason for that pattern may be benign, procedural, data-quality related, or something that merits further review. The method does not decide among those explanations. It produces a hash-bound, caveated investigation queue and stops there.

The leading boundary therefore returns at the end: change-point tests and discrepancy-clustering tests can prioritize investigation. They cannot prove fraud, certify correctness, or replace risk-limiting audits.

6 Conclusion

This paper closes the W-series family survey opened by W.01 by specifying the last two atlas families: change-point tests over reporting sequences and clustering tests over audit discrepancies. Both families fit the same discipline as the rest of the track. They consume hash-bound RCOUNT inputs, emit a common forensic report shape, and attach caveats to every score.

The result is a method contract, not a claim of implementation. Later work may place these analytics in RCOUNT audit-style reports or in a dedicated forensics crate. Whatever the placement, the boundary is fixed: the output is investigative, not certifying; descriptive, not causal; and useful only as a prioritized invitation for further review.

References

Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.

Gio Della-Libera. Election forensic analytics for rcount packages. Technical report, Apportionment Research, 2026b.